

2.6 GRAVITATION P2&P3 QUESTIONS

Compiled by: R. Nkomo

Q1: NOV 2001 P2

- (2) (a) Define gravitational potential ϕ .

.....
[2]

- (b) Fig. 2.1 shows part of the orbit of a satellite round the Earth.

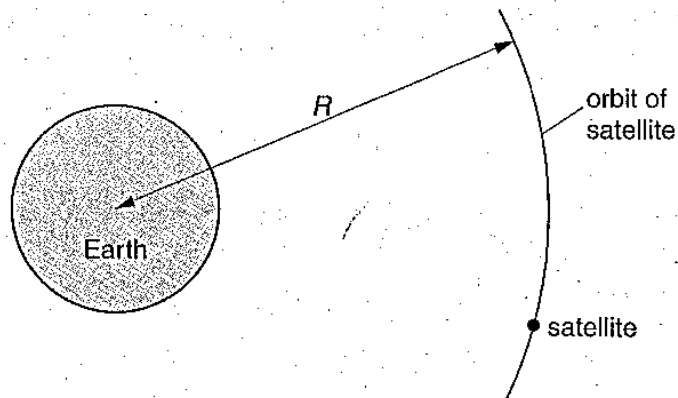


Fig. 2.1

The mass M of the Earth is 6.0×10^{24} kg. It may be assumed that the gravitational field of the Earth is the same as that of a point mass M situated at the centre of the Earth.

- (i) On Fig. 2.1 show, by means of an arrow, the direction of the gravitational force on the satellite.
- (ii) Explain why the satellite does not move in the direction of the gravitational force.

.....

- (iii) Show that v , the linear speed of the satellite in its orbit of radius R , is given by the expression

$$v = \sqrt{\frac{GM}{R}},$$

where G is the gravitational constant.

[5]

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- (c) The satellite is orbiting the Earth with a radius R of 6610 km at a speed v of 7780 m s^{-1} . The satellite is boosted into a higher orbit of radius 6890 km. Show that the speed of the satellite in the new orbit is 7620 m s^{-1} .

[1]

- (d) (i) In (c), the satellite, of mass 120 kg, moves from one orbit to another. Using the data in (c), calculate, for this satellite, the change in

1. kinetic energy,

change in kinetic energy = J

2. gravitational potential energy,

change in potential energy = J

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3. total energy.

change in total energy = J
[6]

(ii) State whether this change in total energy is an increase or a decrease.

.....[1]

Q2:2006 SPEC P2

2 (a) (i) Define *gravitational field strength*.

.....
.....

(ii) State the unit of *gravitational field strength*.

.....
.....
..... [2]

(b) A satellite of mass m orbits a planet of mass M with an orbital radius R .

(i) Taking G to be the universal gravitational constant, write down an expression of the speed v of the satellite in terms of G , M , and R .

.....
.....

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- (ii) Hence or otherwise show that for this satellite $R^3 = \frac{GM}{4\pi^2} T^2$,
where T is the period of the satellite.

[5]

- (c) A moon rotates around jupiter in 8,97 days with an orbital radius of $1,26 \times 10^9$ m. Calculate the mass of jupiter.

mass of jupiter = _____ [2]

Q3:JUN 2011 P2

- 2 (a) State the *universal law of gravitation*.

[2]

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(b) A satellite of mass, m , is in a geostationary orbit.

(i) Define a *geostationary orbit*.

(ii) Show that the radius, r , of the orbit is given by

$$r = \sqrt[3]{\frac{gR^2}{\omega^2}}$$

where symbols have their usual meanings.

[3]

Q4:NOV 2012 P2

3 (a) Explain what is meant by a *geostationary orbit* of a satellite.

[1]

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- (b) A geostationary satellite of mass 120 kg is placed at a distance of 4.4×10^7 m from the centre of the earth.

(i) Calculate the

1. angular velocity, ω , of the satellite,

$$\omega = \underline{\hspace{5cm}}$$

2. speed, v , of the satellite,

$$v = \underline{\hspace{5cm}}$$

3. acceleration, a , of the satellite.

$$a = \underline{\hspace{5cm}}$$

(ii) State **one** use of such a satellite.

[7]

Q5:JUN 2013 P2

- 3 (a) Define *gravitational potential energy*.

[1]

- (b) (i) A man of mass 55.0 kg jumped 1.50 m high on earth.

Deduce

1. the gravitational potential energy, P , gained,

$$P = \underline{\hspace{5cm}}$$

2. the initial velocity, u .

$$u = \underline{\hspace{5cm}} \quad [3]$$

- (ii) The acceleration due to gravity on the moon is $\frac{1}{6}$ that on the earth. The man in (i) jumps with the same initial velocity from the surface of the moon.

Comment on the height reached.

[3]

2.6 GRAVITATION P2&P3 QUESTIONS

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Q6:2018 SPEC P2

- 2 (a) Define *gravitational potential*.

 [2]

- (b) A stone of mass, m , has gravitational potential energy, E_p , at a point, X, in a gravitational field of potential ϕ .

Write an expression for gravitational potential ϕ in terms of m and E_p .

 [1]

- (c) For an isolated spherical planet of radius R , the value of ϕ at its surface is $-6.3 \times 10^7 \text{ Jkg}^{-1}$.

Calculate the change in gravitational potential energy for a 1.4 kg stone moving towards the planet from a distance of $6R$ to $3R$.

Change in gravitational p.e = _____ [3]

2.6 GRAVITATION P2&P3 QUESTIONS

Compiled by: R. Muga

Q7: NOV 2003 P3

- 2 (a) (i) Define *potential* and *field strength* at a point in a gravitational field. [7]
- (ii) State Newton's law of gravitation. Hence, obtain an expression for the field strength due to a point mass at a distance, r , away. [2]
- (iii) Given that the mean radius of the earth is 6.38×10^3 km, deduce the mass of the Earth. [7]
- (b) Calculate the Earth's field strength at a height of 3.3×10^2 km above its surface. [2]
- (c) A satellite orbits the Earth at the height given in (b) above. For the satellite, deduce
- (i) the acceleration,
 - (ii) the linear speed,
 - (iii) the angular speed,
 - (iv) the period. [7]
- (d) Calculate the gain in gravitational potential energy of a satellite of mass 4.25×10^3 kg between its launch and when it is at a height of 3.3×10^2 km above the Earth's surface. [3]
- (e) State one situation under which a body may be described as weightless. [1]

$$v = \sqrt{\frac{GM}{r}}$$

Q8: NOV 2006 P3

- 2 (a) (i) State Newton's law of gravitation. [4]
- (ii) Using the definition of gravitational field strength, show that $g = \frac{GM}{r^2}$ for a point mass m_1 . [4]
- (b) A *geostationary satellite* is at 36 000 km above the earth's surface. The radius of the earth is 6 400 km.
- (i) State what is meant by a *geostationary satellite*. [1]
 - (ii) Calculate the value of the gravitational field strength on the satellite. [4]

2.6 GRAVITATION P2&P3 QUESTIONS

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- (c) A *polar satellite* is one that rotates the earth along the longitudes, as in Fig.2.1

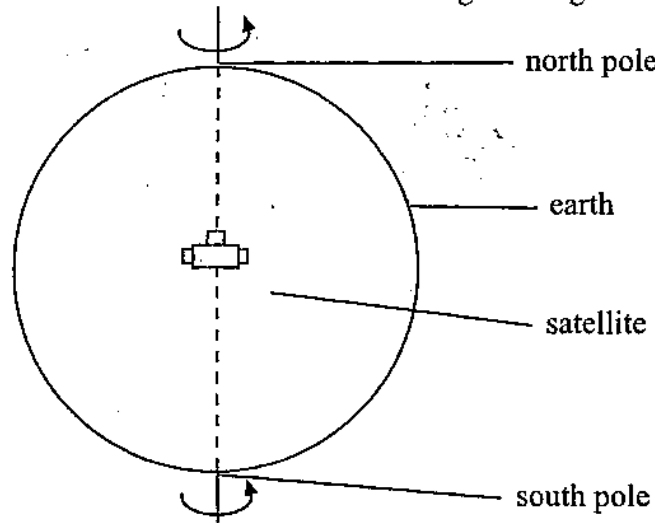


Fig. 2.1

A polar satellite of mass 300 kg has an orbital period of 100 minutes. The mass of the earth is 6.0×10^{24} kg. Calculate

- (i) the height of the satellite above the earth's surface,
 - (ii) the velocity of the satellite. [5]
- (d) Suggest one use for each of the satellites mentioned in (b) and (c). Explain how each use can be achieved, [4]
- (e) The life times of a geostationary and a polar satellite of the same mass are roughly fifteen years and five years respectively. Explain the difference. [3]

Q9:JUN 2007 P3

- 2 (c) A one-tonne satellite is placed in a circular polar orbit so that it goes round the Earth (radius 6.36×10^6 m and mass 5.98×10^{24} kg) 14 times a day.
- (i) Determine the height above the Earth's surface at which it is orbiting.
 - (ii) Calculate the amount of kinetic energy that has to be given to the satellite when it reaches the correct altitude. [7]
- (d) Give a reason for the following:
- (i) the altitude of the satellite decreases
 - (ii) the magnitude of friction increases [2]

Q10:JUN 2008 P3

- 1 (a) The gravitational potential, V , at a point a distance, r , from the centre of the Earth, is given by

$$V = -\frac{GM}{r}, \text{ where } G \text{ and } M \text{ have their usual meanings.}$$

- (i) Explain the significance of the negative sign.
 (ii) Show that the equation is homogeneous. [3]

- (b) A satellite of mass 2 000 kg is placed in a geostationary orbit of radius, R , around the Earth of mass 6.00×10^{24} kg.

- (i) Write down an expression for the gravitational force on the satellite.
 (ii) Explain why the satellite does not fall to the Earth.
 (iii) Show that the period of orbit T of the satellite is given by

$$T = 2\pi \sqrt{\left(\frac{R^3}{4.00 \times 10^{14}} \right)}$$

- (iv) Deduce a value for R , the radius of orbit.
 (v) Calculate the velocity of the satellite in orbit. [10]

- (c) The radius of orbit of the satellite in (b) is increased.

State with a reason what will happen to the

- (i) kinetic energy of the satellite,
 (ii) potential of the satellite,
 (iii) total energy of the satellite. [6]

- (d) Suggest one advantage of geostationary satellites over polar-orbiters. [1]

2.6 GRAVITATION P2&P3 QUESTIONS

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Q11:NOV 2009 P3

- 2 (a) (i) State the universal law of gravitation.
- (ii) Give **one** difference and **one** similarity between this law and Coulomb's law. [4]
- (b) Calculate the work done in taking a 5.0 kg mass from the Earth's surface to a point where the earth's gravitational effect is negligible.
- [Assume that the earth is a uniform sphere of radius 6 400 km and mass 6.0×10^{24} kg.] [3]
- (c) A satellite of mass, m , is placed in an orbit such that it appears stationary above a given point on the Earth.
- Show that the radius of the satellite's orbit is given by
- $r = 574\sqrt[3]{GM_E}$, where M_E is the mass of the Earth and G the universal gravitational constant. [3]

Q12:NOV 2011 P3

- 1 (b) (i) Define *gravitational potential*.
- (ii) Use the universal law of Gravitation to show that the S.I. unit for gravitational constant, G , is $\text{Nm}^2\text{kg}^{-2}$.
- (iii) Explain why
1. inside a planet, the gravitational field falls to zero at the centre,
 2. a satellite in a geostationary orbit, appears to be in a fixed position when viewed from the Earth. [7]

2.6 GRAVITATION P2&P3 QUESTIONS

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Q13:JUN 2012 P3

- 3 (a) Show that the gravitational field strength, g , at any point inside the earth is given by

$$g = 2.8 \times 10^{-10} \rho r,$$

if the earth is assumed to be a perfect sphere of uniform density ρ . (r is the distance from the earth's centre to the point).

[3]

- (b) Determine the value of g at a point 250 km below the earth's surface. (Take the radius of the earth as 6.36×10^6 m).

Suggest why this value is less than the true value at this point.

[3]

- (c) Sketch a graph to show the variation of g with displacement from the earth's centre to a 500 kg satellite that is orbiting the earth at a height of 350 km.

[4]

Q14:JUN 2013 P3

- 1 (c) (i) Define gravitational field strength at a point in a gravitational field.
- (ii) A body of mass 4.0 kg is held 5.0 m above the surface of the Earth.
1. State the gravitational field strength at the position of the body, given that the gravitational field strength near the earth's surface is a constant.
 2. Calculate the gravitational potential at the position of the body.
- (iii) Gravitational potential, $\bar{\phi}$, is a scalar given by

$$\frac{W}{m} = \bar{\phi},$$

where W is the work done in moving a mass, m , from infinity to a point in a gravitational field. Elsewhere the gravitational potential is negative while it is zero at infinity.

Explain why it is negative elsewhere.

[5]

Q15:JUN 2015 P3

- 1 (d) (i)
1. Define the term *gravitational field strength*.
 2. Give an expression for the gravitational field strength, g , at a height, h , above the surface of the earth and explain all the symbols used.
- (ii) An object is dropped from the top of a tall building.
Explain why
1. the gravitational field strength at the top of the building is the same as on the ground,
 2. the acceleration of the object at the top of the building may not be the same as the acceleration near the ground.
- [4]

Q16:NOV 2015 P3

- 4 (a) Define *gravitational field strength*. [1]
- (b) A satellite is in a circular orbit of radius, r , about the earth of mass, M_E .
- (i) Deduce an expression for the period of the satellite.
- (ii) A satellite of mass 500 kg is in circular orbit at a height of 2 000 km above the earth's surface. The radius of the earth is 6 400 km.
- Calculate the
1. gravitational field strength at the height of 2 000 km,
 2. orbital speed of the satellite. [7]
- (c) Suggest why a geostationary satellite has a short lifespan. [2]

Q17: NOV 2016 P3

4 (a) Define the

(i) *gravitational potential* and(ii) *gravitational field strength*.

at a point in a gravitational field.

[2]

(b) (i) State the significance of

1. the negative sign in the equation for gravitational potential,

2. a test mass in the study of the gravitational field.

(ii) Show that the kinetic energy, E_k , of a satellite of mass, M , orbiting the Earth of mass, M_E , at a radius R , is given by $E_k = \frac{GM_E M}{2R}$.

[6]

(c) The satellite in (b)(ii) speeds up when it falls to a lower orbit, hence its kinetic energy increases.

Explain why the satellite loses energy.

[2]

Q18: NOV 2018 P3

1 (e) Satellites A and B, of the same mass are in orbit at heights of 6 370 km and 19 110 km respectively. Radius of the earth is 6 370 km.

Determine the ratio of

(i) potential energy of satellite B to potential energy of satellite A,

(ii) kinetic energy of satellite B to kinetic energy of satellite A.

[4]

Q19: 2018 SPEC P3

1 (d) A satellite of mass m , travels at an angular speed ω , in a circular orbit at a height h , above the surface of planet of mass M and radius R .

(i) State in words Newton's law of universal gravitation.

(ii) Show that the period T , of the satellite is given by

$$T = 2\pi \sqrt{\frac{(R+h)^3}{GM}}$$

(iii) Suggest with a reason, what happens to the speed of the satellite if it moves to an orbit closer to the surface of the planet.

[7]

Q20:NOV 2017 P3

- 4 (a) Define *gravitational field strength*. [1]

- (b) Fig. 4.1 shows a tunnel dug along the diameter of the earth from North to South poles through the centre of the Earth, O.

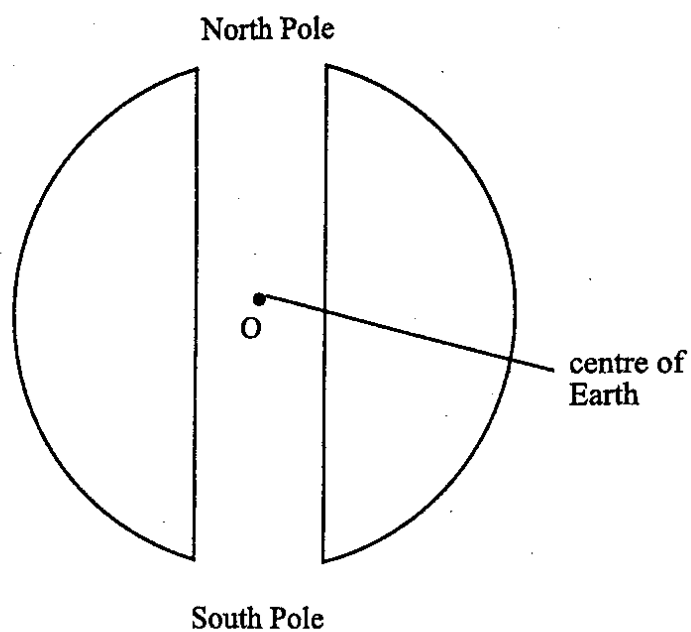


Fig. 4.1

- (i) Using Fig. 4.2 sketch a graph to show how the field strength g , varies through the tunnel.

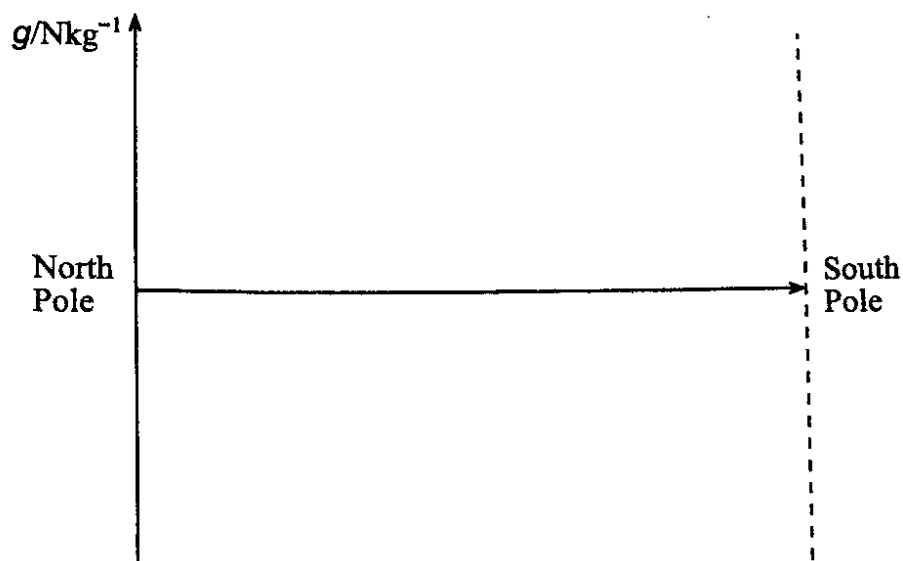


Fig. 4.2

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- (ii) Show that the acceleration a , of a test mass dropped into the tunnel at the North pole is given by $a = -\frac{gx}{R}$ where R is the radius of the Earth and x is any distance from the centre of the Earth.
- (iii) The diameter of the Earth from the North pole to the South pole is 1.28×10^7 m and the field strength on the Earth's surface is 9.81 Nkg^{-1} .

Calculate the time for the test mass to travel from the North pole to South pole through the tunnel.

[7]

Suggest any **two** reasons why a geostationary satellite in orbit has a limited life span.

[2]